Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 9

Time Allowed: 30 Minutes

General Instructions:

Read the following instructions carefully.

1. There are 16 questions in this question paper with internal choice.
2. SECTION A consists of 16 multiple-choice questions carrying 1 mark each.
3. **All questions are compulsory.**
4. **Use of log tables and calculators is not allowed.**

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Section A

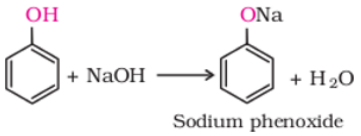
1. Which of the following is a disaccharide? **[1]**

   a) Glucose
   b) Starch
   c) Cellulose
   d) Lactose

2. When diethyl ether is heated with excess of HI, it produces : **[1]**

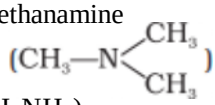
   a) ethyl iodide
   b) ethanol
   c) iodoform
   d) methyl iodide

3. In the reaction **[1]**



   a) Cleavage of O - H bond
   b) Phenols are acidic in nature.
   c) All of these
   d) They can donate a proton to a stronger base

4. Chloromethane on treatment with excess of ammonia yields mainly **[1]**

   A)N, N-Dimethylmethanamine 
   b) N–methylmethanamine (CH_3—NH—CH_3)
   c) Methanamine (CH_3NH_2)
   d) Mixture containing all these in equal proportion

5. Which one among the following metals of 3d series has the lowest melting point? **[1]**

a) Mn
b) Fe
c) Cu
d) Zn

6. Match the item given in Column I with expression given in Column II. **[1]**

Column I	Column II
(a) Osmotic Pressure	(i) $p = K_H . X_B$
(b) Relative lowering of vapour pressure	(ii) $\frac{\Delta P}{P^o_A} = X_B$
(c) Henry Law	(iii) $\Delta T_b = K_b . m$
(d) Elevation in boiling point	(iv) p = iCRT

a) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i).
b) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv).
c) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i).
d) (a) - (iv), (b) - (ii), (c) - (i), (d) - (iii).

7. Which of the following does not give silver mirror test? **[1]**

a) HCOOH
b) CH_3CH_2CHO
c) CH_3COCH_3
d) CH_3CHO

8. The fragrance of the aldehyde and ketone are used for perfume and similar uses depend on: **[1]**

a) moisture of the air.
b) on its reactivity with other functional groups.
c) only solubility of aldehydes and ketones.
d) size and solubility of the aldehyde and ketone molecule.

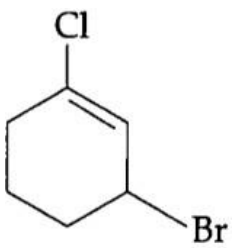
9. For a certain reaction R ⟶ products, a plot of log [R] vs. time gives a straight line with a slope of 1.25 s^{-1}. The order of the reaction is: **[1]**

a) Two
b) One
c) Fractional
d) Zero

10. Value of Henry's constant K_H: **[1]**

a) increases with increase in temperature.
b) increases with decrease in temperature.
c) remains constant.
d) decreases with increase in temperature.

11. The IUPAC name of the compound shown below is: **[1]**



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a) 1-bromo-3-chlorocyclohexene
b) 3-bromo-1-chlorocyclohexene
c) 2-bromo-6-chlorocyclohex-1-ene
d) 6-bromo-2-chlorocyclohexene **[1]**

12. Gabriel synthesis is u12. sed for the preparation of: **[1]**

a) Secondary amine
b) Tertiary amine
c) Quaternary salt
d) Primary amines

13. **Assertion (A):** Vitamin A and D are not absorbed in the body unless fat digestion and absorption proceed normally.

Reason (R): Vitamin A and D are fat-soluble vitamins.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

14. **Assertion (A):** Hydrolysis of ether with aq. HI is S_N1 reaction. **[1]**

Reason (R): $I^{\ominus}$ is strong nucleophile so, it attacks from less hindered side.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.

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c) A is true but R is false. d) A is false but R is true.

15. **Assertion (A):** Rate of addition of HCN on carbonyl compounds increases in presence of NaCN.

a) **Reason (R):** Reaction involves the addition of Cn^- in rate determining step.

b) Both A and R are true and R is the correct explanation of A.Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

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